

SLM Benchmark Report

1. Executive Summary

This report compares the performance of Small Language Models (SLMs) for banking-related tasks. The benchmark covers latency, throughput, and semantic quality.

- Fastest Model: TinyLlama-1.1B-SFT (17.58 tokens/sec)
- Highest Quality: TinyLlama-1.1B-SFT (Similarity: 0.76)

2. Detailed Analysis

The benchmark results highlight a distinct trade-off between inference speed and response quality. 'TinyLlama-1.1B-SFT' leads in throughput, generating 17.58 tokens per second. This speed is advantageous for real-time chat interfaces where low latency improves user satisfaction. However, it achieves a semantic similarity score of 0.76.

Conversely, 'TinyLlama-1.1B-SFT' demonstrates the highest understanding of banking context with a semantic similarity score of 0.76. This higher accuracy ensures that responses align closely with verified banking protocols, which is critical for compliance and trust. The trade-off is a lower throughput of 17.58 tokens per second.

Recommendation: For high-stakes queries (e.g., fraud reporting, fee disputes), prioritize the higher quality model. For general FAQs where speed is essential, the faster model may be sufficient.

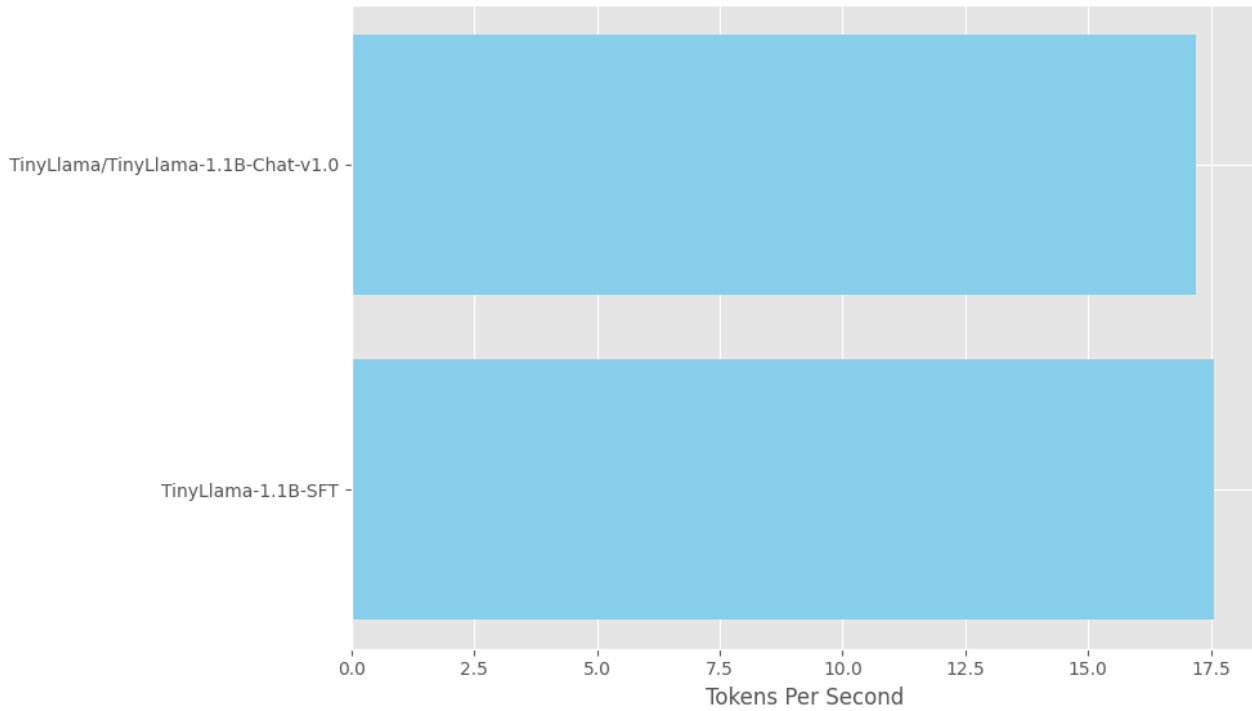
3. Performance Data

Model	Latency (s)	Tokens/s	Similarity
TinyLlama-1.1B-SFT	3.13	17.58	0.76
TinyLlama/TinyLlama-1.1B-Chat-v1.0	8.09	17.21	0.73

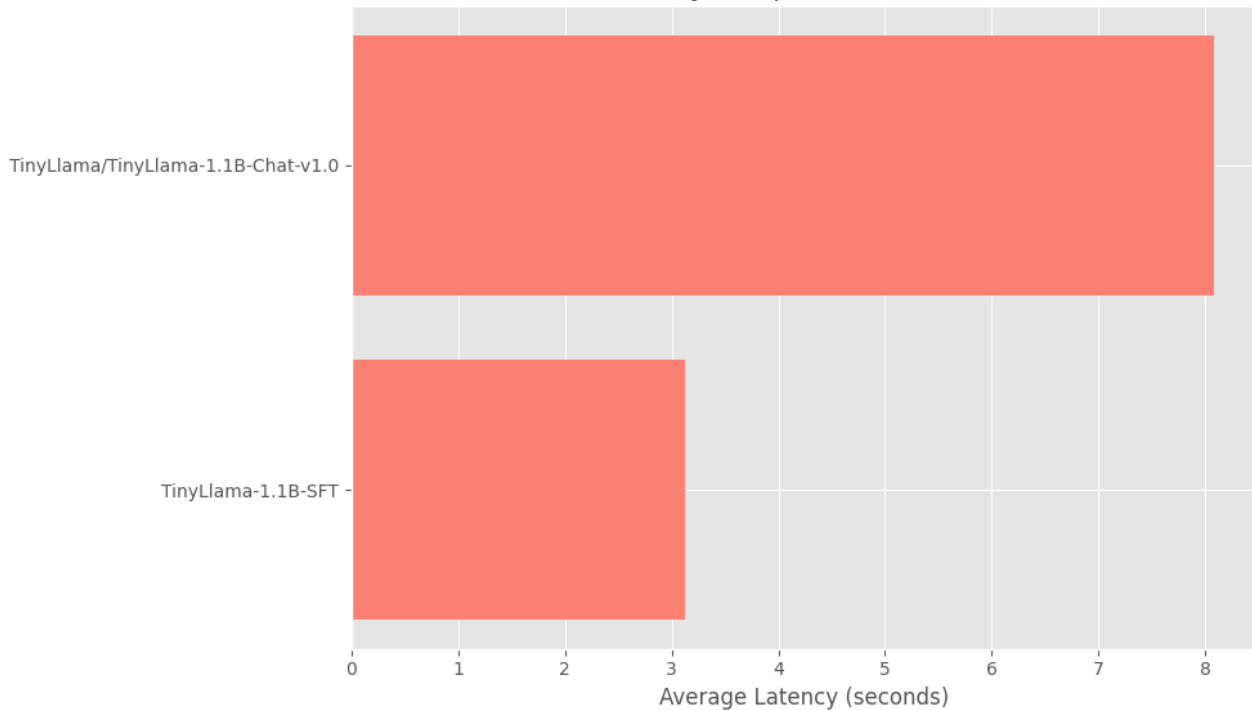
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4. Visual Analysis

Model Throughput Comparison (Higher is Better)



Model Latency Comparison (Lower is Better)



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